Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

If α and $\beta(\alpha<\beta)$ are the roots of the equation $x^2+bx+c=0$, where, $\mathrm{c}<0<\mathrm{b}$, then

(A) $0<\alpha<\beta$

(B) $\alpha<0<\beta<|\alpha|$

(C) $\alpha<\beta<0$

(D) $\alpha<0<|\alpha|<\beta$

Q.2

If the roots of $ax^2+bx+c=0$ are the reciprocals of those of $\ell \mathrm{x}^2+\mathrm{mx}+\mathrm{n}=0$ then $a:b:c=$

(A) $\mathrm{n}:\mathrm{m}:\ell$

(B) $\ell:\mathrm{m}:\mathrm{n}$

(C) $m:n:\ell$

(D) $\mathrm{n}:\ell:\mathrm{m}$

Q.3

The roots of the equation $abc^2x^2+3a^2cx+b^2cx-6a^2-ab+2b^2=0$ are

(A) non real

(B) rational if $\mathrm{a}, \mathrm{b}, \mathrm{c}$, are rational

(C) irrational if a, b, c are rational

(D) None of these

Q.4

If the roots of the equation $x^2-ax+b=0$ are real and differ by a quantity which is less than $\mathrm{c}(\mathrm{c}>0)$, then b lies between

(A) $\frac{a^2-c^2}{4}$ and $\frac{a^2}{4}$

(B) $\frac{a^2+c^2}{4}$ and $\frac{a^2}{4}$

(C) $\frac{a^2-c^2}{2}$ and $\frac{a^2}{4}$

(D) None of these

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Q.5

For the equation $\left|x^2\right| + |x| - 6 = 0$, the roots are

(A) One and only one real number

(B) Real with sum one

(C) Real with sum zero

(D) Real with product zero

Q.6

If the roots of the equation $ax^2 - bx + c = 0$ are α, then the roots are α and β of the equation $b^2cx^2 - ab^2x + a^3 = 0$ are

(A) $\frac{1}{\alpha^3+\alpha\beta}, \frac{1}{\beta^3+\alpha\beta}$

(B) $\frac{1}{\alpha^2+\alpha\beta}, \frac{1}{\beta^2+\alpha\beta}$

(C) $\frac{1}{\alpha^4+\alpha\beta}, \frac{1}{\beta^4+\alpha\beta}$

(D) None of these

Q.7

If α, β be the roots of the equation $x^2 - px + q = 0$, then
the equation whose roots are
$\alpha^2\left(\frac{\alpha^2}{\beta} - \beta\right)$ and $\beta^2\left(\frac{\beta^2}{\alpha} - \alpha\right)$ is

(A) $qx^2 - p\left(p^2 - q\right)\left(p^2 - 4q\right)x - p^2q^2\left(p^2 - 4q\right) = 0$

(B) $px^2 - q\left(p^2 - p\right)\left(p^2 - 4q\right)x + p^2q^2\left(p^2 - 4q\right) = 0$

(C) $px^2 - qx + p = 0$

(D) None of these

Q.8

Let $a > 0, b > 0$ and $c > 0$. Then both the roots of the equation $ax^2 + bx + c = 0$

(A) are real and negative

(B) have negative real parts

(C) are rational numbers

(D) None of these

Q.9

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If $(7-4\sqrt{3})^{x^2-4x+3}+(7+4\sqrt{3})^{x^2-4x+3}=14$, then the value of x is given by

(A) $2, 2\pm\sqrt{2}$

(B) $2\pm\sqrt{3}, 3$

(C) $3\pm\sqrt{2}, 2$

(D) None of these

Q.10

If a, b, c are real numbers $a \neq 0$. If α, is a root of $a^2x^2+bx+c=0$, β is a root of $a^2x^2-bx-c=0$ and $0<\alpha<\beta$, then the equation $a^2x^2+2bx+2c=0$ has a γ root that always satisfies:

(A) $\gamma=\frac{\alpha+\beta}{2}$

(B) $\gamma=\frac{\alpha-\beta}{2}$

(C) $\gamma=\alpha$

(D) $\alpha<\gamma<\beta$

Q.11

If the roots of the equation $(x-a)(x-b)+(x-b)(x-c)+(x-c)(x-a)=0$ are equal, then $a^2+b^2+c^2=$

(A) $a+b+c$

(B) $2a+b+c$

(C) $3abc$

(D) $ab+bc+ca$

Q.12

If α, β are the roots of the equation $ax^2+bx+c=0$ such that $\beta<\alpha<0$, then the quadratic equation whose roots are $|\alpha|, |\beta|$, is given by

(A) $|a|x^2+|b|x+|c|=0$

(B) $ax^2-|b|x+c=0$

(C) $|a|x^2-|b|x+|c|=0$

(D) $a|x|^2+b|x|+|c|=0$

Q.13

The equation $(\cos p-1)x^2+(\cos p)x+\sin p=0$, where x is a variable, has real roots if p lies in the interval

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(A) $(0, 2\pi)$

(B) $(-\pi, 0)$

(C) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

(D) $[0, \pi]$

Q.14

If the product of the roots of the equation $x^2 - 5kx + 2e^{4\ln k} - 1 = 0$ is 31 , then sum of the root is

(A) -10

(B) 5

(C) -8

(D) none of these

Q.15

If $\sin\theta$ and $\cos\theta$ are the roots of the equation $ax^2 - bx + c = 0$, then a, b and c satisfy the relation

(A) $a^2 + b^2 + 2ac = 0$

(B) $a^2 - b^2 + 2ac = 0$

(C) $a^2 + c^2 + 2ab = 0$

(D) $a^2 - b^2 - 2ac = 0$

Q.16

The set of all real numbers x for which $x^2 - |x + 2| + x > 0$, is

(A) $(-\infty, -2) \cup (2, \infty)$

(B) $(-\infty, -\sqrt{2}) \cup (\sqrt{2}, \infty)$

(C) $(-\infty, -1) \cup (1, \infty)$

(D) $(\sqrt{2}, \infty)$

Q.17

If x be real and $b < c$, then $\frac{x^2 - bc}{2x - b - c}$ lies in

(A) (b, c)

(B) $[\mathrm{b}, \mathrm{c}]$

(C) $(-\infty, b] \cup [c, \infty)$

(D) $(-\infty, b) \cup (c, \infty)$

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Q.18

If both the roots of the equation $x^2 - 2kx + k^2 - 4 = 0$ lie between -3 and 5, then which one of the following is correct?

(A) $-2 < \mathrm{k} < 2$

(B) $-5 < \mathrm{k} < 3$

(C) $-3 < \mathrm{k} < 5$

(D) $-1 < \mathrm{k} < 3$

Q.19

If a and b are the non-zero distinct roots of $x^2 + ax + b = 0$, then the least value of $x^2 + ax + b$ is

(A) $\frac{2}{3}$

(B) $\frac{9}{4}$

(C) $-\frac{9}{4}$

(D) 1

Q.20

The expression $\frac{x^2-2x+p^2}{x^2+2x+p^2}$, where $p > 1$, lies in between

(A) $\frac{p-1}{p+1}$ and $\frac{p+1}{p-1}$

(B) $\frac{p-1}{p+1}$ and

(C) $p^2 - 1$ and $p^2 + 1$

(D) p and p^2

Answers & Solutions

Q.1 Answer:

$\alpha < 0 < \beta < |\alpha|$

Solution:

Given $\alpha < \beta, \mathrm{c} < 0, \ \mathrm{b} > 0,$

$\therefore \alpha + \beta = -\mathrm{b} < 0$ **and** $\alpha\beta = \mathrm{c} < 0$

Clearly, α **and** β **have opposite signs and** $\alpha < \beta$

$\therefore \alpha < 0$ **and** $\beta > 0 \Rightarrow \alpha < 0 < \beta$

Further $\alpha + \beta < 0 \Rightarrow \beta < -\alpha \Rightarrow |\beta| < |-\alpha|$

$\Rightarrow \beta < |\alpha| (\beta > 0 \Rightarrow |\beta| = \beta)$

Hence, $\alpha < 0 < \beta < |\alpha|$

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Q.2 Answer:

$\mathrm{n} : \mathrm{m} : \ell$

Solution:

If α, β **be the roots then** $\alpha+\beta=-\frac{\mathrm{b}}{\mathrm{a}}, \alpha\beta=\frac{\mathrm{c}}{\mathrm{a}}$
Now the roots of $\ell \mathrm{x}^2+\mathrm{mx}+\mathrm{n}=0$ **are** $\frac{1}{\alpha}, \frac{1}{\beta}$
$\therefore \frac{1}{\alpha}+\frac{1}{\beta}=-\frac{\mathrm{m}}{\ell}$ **and** $\frac{1}{\alpha} \cdot \frac{1}{\beta}=\frac{\mathrm{n}}{\ell}$
or $\frac{\alpha+\beta}{\alpha\beta}=-\frac{\mathrm{m}}{\ell}$ **and** $\frac{\mathrm{a}}{\mathrm{c}}=\frac{\mathrm{n}}{\ell}$
or $-\frac{b}{c}=-\frac{m}{\ell}$ **and** $\frac{a}{c}=\frac{n}{\ell}$
or $\quad \frac{a}{n}=\frac{b}{m}=\frac{c}{\ell} . \therefore a : b : c=n : m : \ell$

Q.3 Answer:

rational if $\mathrm{a}, \mathrm{b}, \mathrm{c}$, are rational

Solution:

The equation is $abc^2x^2+\left(3a^2c+b^2c\right)x-6a^2-ab+2b^2=0$
Discriminant
$D=\left(3a^2+b^2\right)^2c^2-4abc^2\left(-6a^2-ab+2b^2\right)$
$=9a^4c^2+b^4c^2+6a^2b^2c^2+24a^3bc^2$
$+4a^2b^2c^2-8ab^3c^2$
$=9a^4c^2+16a^2b^2c^2+b^4c^2+24a^3bc^2$
$-8ab^3c^2-6a^2b^2c^2$
$=\left(3a^2c+4abc-b^2c\right)^2$
Since, the discriminant is a prefect square, therefore the roots are rational provided $\mathrm{a}, \mathrm{b}, \mathrm{c}$, **are rational.**

Q.4 Answer:

$\frac{a^2-c^2}{4}$ and $\frac{a^2}{4}$

Solution:

Given roots are real and distinct, then $a^2-4b>0$
$\Rightarrow \quad b<a^2/4$
Again α **and** β **differ by a quantity less than** $c(c>0) \Rightarrow|\alpha-\beta|<\operatorname{cor}(\alpha-\beta)^2<c^2$
$\Rightarrow \quad (\alpha+\beta)^2-4\alpha\beta<c^2$ **or** $a^2-4b<c^2$
or $\frac{a^2-c^2}{4}<b$
$\Rightarrow \quad \frac{a^2-c^2}{4}<b<\frac{a^2}{4}$ **by** (1) **and** (2)

Q.5 Answer:

Real with sum zero

Solution:

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When $x < 0, |x| = -x$

$\therefore$ **Equation is** $x^2 - x - 6 = 0 \Rightarrow x = -2, 3$

$\because$ $x < 0, \therefore x = -2$ **is the solution.**

When $x \geq 0, |x| = x$

$\therefore$ **Equation is** $x^2 + x - 6 = 0 \Rightarrow x = 2, -3$

$\because$ $x \geq 0, \therefore x = 2$ **is the solution,**

Hence $x = 2, -2$ **are the solutions and their sum is zero.**

Q.6 Answer:

$\frac{1}{\alpha^2+\alpha\beta}, \frac{1}{\beta^2+\alpha\beta}$

Solution:

Multiplying the second equation by $\frac{c}{a^3}$

we get $\frac{b^2c^2}{a^3}x^2 - \frac{b^2c}{a^2}x + c = 0$

$\Rightarrow \quad a\left(\frac{bc}{a^2}x\right)^2 - b\left(\frac{bc}{a^2}\right)x + c = 0$

$\Rightarrow \frac{bc}{a^2}x = \alpha, \beta$

$\Rightarrow \quad (\alpha+\beta)\alpha\beta x = \alpha, \beta$

$\Rightarrow x = \frac{1}{(\alpha+\beta)\alpha}, \frac{1}{(\alpha+\beta)\beta}$

Q.7 Answer:

$qx^2 - p\left(p^2 - q\right)\left(p^2 - 4q\right)x - p^2q^2\left(p^2 - 4q\right) = 0$

Solution:

Here, $\alpha + \beta = \mathrm{p}$ **and** $\alpha\beta = \mathrm{q}$

Let $\alpha_1 = \alpha^2\left(\frac{\alpha^2}{\beta} - \beta\right)$ **and** $\beta_1 = \beta^2\left(\frac{\beta^2}{\alpha} - \alpha\right)$ **then**

Sum of the roots $\alpha_1 + \beta_1 = \alpha^2\left(\frac{\alpha^2}{\beta} - \beta\right) + \beta^2\left(\frac{\beta^2}{\alpha} - \alpha\right)$

$= \left(\alpha^2 - \beta^2\right)\left(\frac{\alpha^2}{\beta} - \frac{\beta^2}{\alpha}\right) = \frac{1}{\alpha\beta}(\alpha - \beta)(\alpha + \beta)\left(\alpha^3 - \beta^3\right)$

$= \frac{1}{\alpha\beta}(\alpha + \beta)(\alpha - \beta)^2\left(\alpha^2 + \beta^2 + \alpha\beta\right)$

$= \frac{1}{\alpha\beta}(\alpha + \beta)\left\{(\alpha + \beta)^2 - 4\alpha\beta\right\}\left((\alpha + \beta)^2 - \alpha\beta\right\}$

$= \frac{1}{q}(p)\left\{p^2 - 4q\right\}\left\{p^2 - q\right\}$

Product of the roots $\alpha_1\beta_1 = \alpha^2\left(\frac{\alpha^2}{\beta} - \beta\right) \cdot \beta^2\left(\frac{\beta^2}{\alpha} - \alpha\right)$

$= -\alpha\beta\left(\alpha^2 - \beta^2\right)^2 = -\alpha\beta(\alpha + \beta)^2(\alpha - \beta)^2$

$= -\alpha\beta(\alpha + \beta)^2\left[(\alpha + \beta)^2 - 4\alpha\beta\right] = -\mathrm{p}^2\mathrm{q}\left(\mathrm{p}^2 - 4\mathrm{q}\right)$

$\therefore$ **The required equation is** $x^2 - \frac{p}{q}\left(p^2 - 4q\right)\left(p^2 - q\right)x - p^2q\left(p^2 - 4q\right) = 0$

or $\mathrm{qx}^2 - \mathrm{p}\left(\mathrm{p}^2 - \mathrm{q}\right)\left(\mathrm{p}^2 - 4\mathrm{q}\right)\mathrm{x} - \mathrm{p}^2\mathrm{q}^2\left(\mathrm{p}^2 - 4\mathrm{q}\right) = 0$

Q.8 Answer:

have negative real parts

Solution:

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Since, $a, b, c > 0$ and $ax^2 + bx + c = 0$ then,

$\Rightarrow x = \frac{-b}{2a} \pm \frac{\sqrt{b^2-4ac}}{2a}$

Case 1 : When $b^2 - 4ac > 0 \Rightarrow x = \frac{-b}{2a} - \frac{\sqrt{b^2-4ac}}{2a}$ and $\frac{-b}{2a} + \frac{\sqrt{b^2-4ac}}{2a}$ both roots are negative.

Case 2 : Ehen $b^2 - 4ac = 0 \Rightarrow x = \frac{-b}{2a}$ i.e., both roots are equal and negative.

Case 3 : When $b^2 - 4ac < 0 \Rightarrow x = \frac{-b}{2a} \pm i\frac{\sqrt{4ac-b^2}}{2a}$ have negative real part.

$\therefore$ From above discission both roots have negative real parts.

Q.9 Answer:

$2, 2 \pm \sqrt{2}$

Solution:

Since $(7 + 4\sqrt{3})(7 - 4\sqrt{3}) = 1,$

$\therefore$ The given equation becomes

$y + \frac{1}{y} = 14$ where $y = (7 - 4\sqrt{3})^{x^2-4x+3}$

$\Rightarrow y^2 - 14y + 1 = 0 \Rightarrow y = 7 \pm 4\sqrt{3}$

Now $y = 7 + 4\sqrt{3} \Rightarrow x^2 - 4x + 3 = -1 \Rightarrow x = 2$

Also $y = 7 - 4\sqrt{3} \Rightarrow x^2 - 4x + 3 = 1 \Rightarrow x = 2 \pm \sqrt{2}$

Q.10 Answer:

$\alpha < \gamma < \beta$

Solution:

α is a root of $a^2x^2 + bx + c = 0 \Rightarrow a^2\alpha^2 + b\alpha + c = 0$

$\Rightarrow \quad b\alpha + c = -a^2\alpha^2$ ...(i)

β is a roots of $a^2x^2 - bx - c = 0 \Rightarrow b\beta + c = a^2\beta^2$ ...(ii)

Let $f(x) = a^2x^2 + 2bx + 2c$. Then,

$f(\alpha) = a^2\alpha^2 + 2b\alpha + 2c = a^2\alpha^2 + 2(b\alpha + c)$

$\Rightarrow f(\alpha) = a^2\alpha^2 - 2a^2\alpha^2 \quad [\text{Using (i)}]$

$= -a^2\alpha^2 < 0$

and $f(\beta) = a^2\beta^2 + 2b\beta + 2c$

$= a^2\beta^2 + 2(b\beta + c)$

$= a^2\beta^2 + 2a^2\beta^2 = 3a^2\beta^2 > 0$ [Using (ii)]

Thus, $f(x)$ is a polynomial such that $f(\alpha) < 0$ and $f(\beta) > 0$. Therefore, there exists γ satisfying $\alpha < \gamma < \beta$ such that $f(\gamma) = 0$

Q.11 Answer:

$ab + bc + ca$

Solution:

Given equation is $(x - a)(x - b) + (x - b)(x - c) + (x - c)(x - a) = 0$

$\Rightarrow (x^2 - bx - ax + ab) + (x^2 - cx - bx + bc)$

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$+\left(x^2-ax-cx+ac\right)=0$
$\Rightarrow \left[x^2-(a+b)x+ab\right]+\left[x^2-(b+c)x+bc\right]$
$+\left[x^2-(a+c)x+ac\right]=0$
$\Rightarrow 3x^2-2(a+b+c)x+ab+bc+ca=0$
Since roots are equal We have $b^2-4ac=0\Rightarrow b^2=4ac$
$4(a+b+c)^2=12(ab+bc+ca)$
$(a+b+c)^2=3(ab+bc+ca)$
$a^2+b^2+c^2=ab+bc+ca$

Q.12 Answer:

$|a|x^2-|b|x+|c|=0$

Solution:

Since, α **and** β **be the roots of the equation** $ax^2+bx+c=0$**, then**

$$\begin{aligned}&\alpha+\beta=-\tfrac{b}{a} \text{ and } \alpha\beta=\tfrac{c}{a}\\ &\text{Now, sum of roots } =|\alpha|+|\beta|\\ &=-\alpha-\beta \qquad (\because \quad \beta<\alpha<0)\\ &=-\left(-\tfrac{b}{a}\right)=\left|\tfrac{b}{a}\right| \quad (\because \quad |\alpha|+|\beta|>0)\end{aligned}$$

and product of roots $=|\alpha||\beta|=\left|\frac{c}{a}\right|$ **Hence, required equation is**

$$\begin{aligned}&x^2-\left|\frac{b}{a}\right|x+\left|\frac{c}{a}\right|=0\\ \Rightarrow \quad &|a|x^2-|b|x+|c|=0\end{aligned}$$

Q.13 Answer:

$[0,\pi]$

Solution:

The equation has real roots if
$\cos^2 p-4(\cos p-1)\sin p\geq 0$ **or** $\cos^2 p+4\sin p(1-\cos p)\geq 0$
which will be necessarily true if $\sin p\geq 0$ **i.e.** $0\leq p\leq \pi$

Q.14 Answer:

none of these

Solution:

We have product to the roots $=2e^{4\,\mathrm{m}k}-1=31$ **(given)**
$\Rightarrow \quad e^{4\ln k}=16\Rightarrow e^{\mathrm{mn}\left(k^4\right)}=16\Rightarrow k^4=16\Rightarrow k^4-16=0$
$\Rightarrow \quad (k-2)(k+2)\left(k^2+4\right)=0\Rightarrow k=2,-2$

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As $k > 0$, we get $k = 2$

Sum of the roots $= 5k = 10$

Q.15 Answer:

$a^2 - b^2 + 2ac = 0$

Solution:

Since $\sin\theta$ and $\cos\theta$ are roots of the given quadratic equation, we have

$\sin\theta + \cos\theta = \frac{b}{a}$ and $\sin\theta\cos\theta = \frac{c}{a}$

$\Rightarrow \quad (\sin\theta + \cos\theta)^2 = \frac{b^2}{a^2} \Rightarrow \sin^2\theta + \cos^2\theta + 2\sin\theta\cos\theta = \frac{b^2}{a^2}$

$\Rightarrow 1 + 2\frac{c}{a} = \frac{b^2}{a^2} \Rightarrow a^2 + 2ac - b^2 = 0$

Q.16 Answer:

$(-\infty, -\sqrt{2}) \cup (\sqrt{2}, \infty)$

Solution:

For $x \geq -2, x^2 - x - 2 + x > 0$

$\Rightarrow \mathrm{x}^2 > 2 \Rightarrow \mathrm{x} \in (-\infty, -\sqrt{2}) \cup (\sqrt{2}, \infty)$

$\Rightarrow x \in [-2, -\sqrt{2}) \cup (\sqrt{2}, \infty)$ For $x < -2$

$x^2 + x + 2 + x > 0$ or $x^2 + 2x + 2 > 0$

which is true for all x. Hence $x \in (-\infty, -\sqrt{2}) \cup (\sqrt{2}, \infty)$

Q.17 Answer:

$(-\infty, b] \cup [c, \infty)$

Solution:

Let $y = \frac{x^2 - bc}{2x - b - c}$

$\Rightarrow \mathrm{x}^2 - 2\mathrm{yx} + (\mathrm{b} + \mathrm{c})\mathrm{y} - \mathrm{bc} = 0$

$\because \mathrm{x} \in \mathrm{R}$, so $4\mathrm{y}^2 - 4(\mathrm{b} + \mathrm{c})\mathrm{y} + 4\mathrm{bc} \geq 0$

$\Rightarrow \quad x \leq b$ or $x \geq c \quad (\because b < c)$

Q.18 Answer:

$-1 < \mathrm{k} < 3$

Solution:

$x^2 - 2kx + k^2 - 4 = 0$

$\Rightarrow \quad (\mathrm{x} - \mathrm{k})^2 - 2^2 = 0$

$\Rightarrow \quad (x - k - 2)(x - k + 2) = 0$

$\Rightarrow \quad \mathrm{x} = \mathrm{k} + 2, \mathrm{k} - 2$

$\Rightarrow \quad \mathrm{k} + 2 < 5 \& \mathrm{k} - 2 > -3$

$\Rightarrow \quad k < 3 \& k > -1$

$\Rightarrow -1 < \mathrm{k} < 3$

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Q.19 Answer:

$-\frac{9}{4}$

Solution:

Given $a+b=-a, ab=b$

As $b \neq 0$**, we get** $a=1$

$\therefore \quad 1+b=-1 \Rightarrow b=-2$

Thus, $x^2+ax+b=x^2+x-2=\left(x+\frac{1}{2}\right)^2-\frac{9}{4} \geq -\frac{9}{4}$

$\therefore$ **Least value of** x^2+ax+b **is** $-\frac{9}{4}$ **which is attained at** $x=-\frac{1}{2}$

Q.20 Answer:

$\frac{p-1}{p+1}$ and $\frac{p+1}{p-1}$

Solution:

Let $y=\frac{x^2-2x+p^2}{x^2+2x+p^2}$

$\therefore yx^2+2yx+p^2y-x^2+2x-p^2=0$

$\Rightarrow (y-1)x^2+(2y+2)x+p^2y-p^2=0$

since x **is real, we have,** $(2y+2)^2-4(y-1)\left(p^2y-p^2\right) \geq 0 \Rightarrow p^2(y-1)^2-(y+1)^2 \leq 0 \quad (\because p>1)$

$\Rightarrow \quad y^2-\frac{2p^2+2}{p^2-1}y+1 \leq 0 \Rightarrow y^2-\left(\frac{p-1}{p+1}+\frac{p+1}{p-1}\right)y+\left(\frac{p-1}{p+1}\right)\left(\frac{p+1}{p-1}\right) \leq 0$

$\Rightarrow \left(y-\frac{p-1}{p+1}\right)\left(y-\frac{p+1}{p-1}\right) \leq 0 \quad \Rightarrow \frac{p-1}{p+1} \leq y \leq \frac{p+1}{p-1}$

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